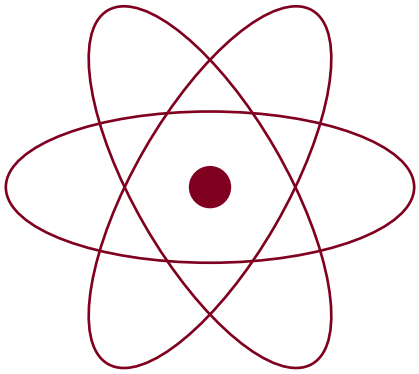
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SAUDI PHYSICS OLYMPIAD HUB



Kinematics I – Solutions

Detailed Olympiad Solutions

Olympiad Training • Mechanics

SPhOH Team

SECOND EDITION • 2026

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Preface

This document provides complete, step-by-step mathematical solutions for all 21 problems presented in the **Kinematics I** problem set published by the **Saudi Physics Olympiad Hub (SPhOH)**.

Rigorous analytical derivations, coordinate transformations, relative motion frame choices, and physical explanations are detailed for each question.

CHAPTER 1

Kinematics I Solutions

1.1 1D Kinematics and Kinematic Equations

SOLUTION 1 — LEVEL I

Acceleration along a Straight Line

We are given the initial velocity $v_0 = 1$ m/s, final velocity $v = 14$ m/s, and displacement $\Delta x = 20$ m.
Using the 1D kinematic relationship:

$$v^2 = v_0^2 + 2a\Delta x$$

Substitute the known values into the equation:

$$(14)^2 = (1)^2 + 2a(20) \implies 196 = 1 + 40a$$

Solving for the acceleration a:

$$40a = 195 \implies a = \frac{195}{40} = 4.875 \text{ m/s}^2$$

The acceleration has a magnitude of 4.875 m/s^2 (or 4.88 m/s^2) directed along the direction of motion (positive x-axis).

SOLUTION 2 — LEVEL II

Multi-Stage Average Velocity

The motion consists of two distinct stages:

- **Stage 1:** Distance $\Delta x_1 = 30$ at constant speed $v_1 = 8$ 1/h.

$$\Delta t_1 = \frac{\Delta x_1}{v_1} = \frac{30 \text{ km}}{8 \text{ km/h}} = 3.75 \text{ h}$$

- **Stage 2:** Distance $\Delta x_2 = 18$ at constant speed $v_2 = 10$ 1/h.

$$\Delta t_2 = \frac{\Delta x_2}{v_2} = \frac{18 \text{ km}}{10 \text{ km/h}} = 1.80 \text{ h}$$

Average speed over the total journey is given by total distance divided by total elapsed time:

$$v_{\text{avg}} = \frac{\Delta x_{\text{total}}}{\Delta t_{\text{total}}} = \frac{\Delta x_1 + \Delta x_2}{\Delta t_1 + \Delta t_2} = \frac{30 + 18}{3.75 + 1.80} = \frac{48 \text{ km}}{5.55 \text{ h}} \approx 8.65 \text{ 1/h}$$

SOLUTION 3

LEVEL II

Average Velocity over Split Distance and Time

Let the total displacement be S. The journey is split into two halves of distance $S/2$:
First Half ($S/2$): Moving with constant velocity v_0. The time spent is:

$$t_1 = \frac{S/2}{v_0} = \frac{S}{2v_0}$$

Second Half ($S/2$): Let the total time taken for this half be T_2. It is split into two equal time intervals $\tau = T_2/2$:

- Distance covered in first half of time: $s_a = v_1\tau = v_1\left(\frac{T_2}{2}\right)$
- Distance covered in second half of time: $s_b = v_2\tau = v_2\left(\frac{T_2}{2}\right)$

The total distance for the second half is $S/2$:

$$\frac{S}{2} = s_a + s_b = \frac{T_2}{2}(v_1 + v_2) \implies T_2 = \frac{S}{v_1 + v_2}$$

Thus, the effective average speed v' over the second distance $S/2$ is:

$$v' = \frac{S/2}{T_2} = \frac{v_1 + v_2}{2}$$

Entire Journey: Combining the two equal-distance halves ($S/2$ and $S/2$):

$$v_{\text{avg}} = \frac{S}{t_1 + T_2} = \frac{S}{\frac{S}{2v_0} + \frac{S}{v_1+v_2}} = \frac{1}{\frac{1}{2v_0} + \frac{1}{v_1+v_2}} = \frac{2v_0(v_1 + v_2)}{2v_0 + v_1 + v_2}$$

SOLUTION 4

LEVEL II

Motorboat and Floating Raft

Consider the problem in the reference frame of the river current.
In this frame, the water is at rest, meaning the raft (which drifts passively with the current) remains stationary at point A. The motorboat moves relative to the water at a constant speed v_{boat}.

1. Moving away from the raft, the boat travels for time $\tau = 60$ min.
2. Turning back toward the raft at the exact same power setting, the boat moves relative to the still water at speed v_{boat}. Because the distance relative to the raft is identical, the boat takes the exact same time $\tau = 60$ min to return to the raft.

Therefore, the total time elapsed from initial separation to reunion is:

$$t_{\text{total}} = 2\tau = 2 \times 60 \text{ min} = 120 \text{ min} = 2.0 \text{ h}$$

During this time t_{total}, the raft has drifted a distance $l = 6.0$ in the ground frame. Thus, the river flow speed is:

$$v_{\text{flow}} = \frac{l}{t_{\text{total}}} = \frac{6.0 \text{ km}}{2.0 \text{ h}} = 3.0 \text{ 1/h}$$

SOLUTION 5 LEVEL II

2D Motion under Constant Acceleration

Given $\vec{r}_0 = 3\hat{i} - 7\hat{j}$, $\vec{v}_0 = 2\hat{i}$, and $\vec{a} = \hat{i} + 3\hat{j}$. The position vector as a function of time is:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2 = \left(3 + 2t + \frac{1}{2}t^2\right)\hat{i} + \left(-7 + \frac{3}{2}t^2\right)\hat{j}$$

(a) Crossing the x-axis ($y = 0$):

$$y(t) = -7 + \frac{3}{2}t^2 = 0 \implies t^2 = \frac{14}{3} \implies t = \sqrt{\frac{14}{3}} \approx 2.16 \text{ s}$$

Substituting $t = \sqrt{14/3}$ into $x(t)$:

$$x = 3 + 2\left(\sqrt{\frac{14}{3}}\right) + \frac{1}{2}\left(\frac{14}{3}\right) = 3 + \frac{7}{3} + 2\sqrt{\frac{14}{3}} = \frac{16}{3} + 2\sqrt{\frac{14}{3}} \approx 9.65 \text{ m}$$

(b) Crossing the y-axis ($x = 0$):

$$x(t) = 3 + 2t + \frac{1}{2}t^2 = 0 \implies t^2 + 4t + 6 = 0$$

The discriminant of this quadratic equation is $\Delta = 4^2 - 4(1)(6) = -8 < 0$. Since there are no real roots for t, the particle **never crosses the y-axis**.

1.2 Vector Displacement in 3D Space

SOLUTION 6 **LEVEL II**

3D Box Corner-to-Corner Fly Displacement

The fly moves from corner $A(0, 0, 0)$ to diagonally opposite corner $B(L, W, H)$.
The displacement vector $\Delta\vec{r}$ connecting point A to point B in Cartesian coordinates is:

$$\Delta\vec{r} = (L - 0)\hat{i} + (W - 0)\hat{j} + (H - 0)\hat{k} = L\hat{i} + W\hat{j} + H\hat{k}$$

The magnitude of the displacement vector is calculated via the 3D Pythagorean theorem:

$$|\Delta\vec{r}| = \sqrt{L^2 + W^2 + H^2}$$

1.3 Graphical Analysis of Motion

SOLUTION 7 **LEVEL I**

Position from Velocity–Time Graph

From the linear graph, $v(t) = 0$ occurs at $t = 3$ s. We are given $x(3) = 4$ m.
The velocity function is linear: $v(t) = 3 - t$ m/s. The displacement $\Delta x = x(3) - x(0)$ is the area under the $v(t)$ curve from $t = 0$ to $t = 3$ s:

$$\Delta x = \text{Area of Triangle} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times (3\text{ s}) \times (3\text{ m/s}) = 4.5\text{ m}$$

By definition of displacement:

$$x(3) - x(0) = 4.5 \implies 4 - x(0) = 4.5 \implies x(0) = -0.5\text{ m}$$

The initial position of the object is $x_0 = -0.5$ m.

SOLUTION 8 **LEVEL I**

Acceleration from Velocity–Time Graph

Instantaneous acceleration is given by the slope of the $v(t)$ graph:

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t}$$

The line segment covering $t = 0$ passes through the points $(0\ \mathrm{s}, -2\ \mathrm{m/s})$ and $(2\ \mathrm{s}, 4\ \mathrm{m/s})$. Calculating the slope:

$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{4 - (-2)}{2 - 0} = \frac{6}{2} = 3\ \mathrm{m/s^2}$$

The instantaneous acceleration at $t = 0$ is $3\ \mathrm{m/s^2}$.

SOLUTION 9 **LEVEL I**

Instantaneous Velocity from Position–Time Graph

Instantaneous velocity is given by the slope of the position–time graph:

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t}$$

From the graph, the line passes through $(0\ \mathrm{s}, 6\ \mathrm{m})$ and $(3\ \mathrm{s}, 0\ \mathrm{m})$.
Calculating the slope:

$$v = \frac{x_2 - x_1}{t_2 - t_1} = \frac{0 - 6}{3 - 0} = -2\ \mathrm{m/s}$$

The instantaneous velocity at $t = 0$ is $-2\ \mathrm{m/s}$.

1.4 Free Fall and Vertical Motion under Gravity

SOLUTION 10 **LEVEL III**

Continuous Drop of Falling Balls

(a) **First ball released from rest:** Distance dropped in first interval t_0: $h = \frac{1}{2}gt_0^2$. Total distance dropped after time $2t_0$: $y(2t_0) = \frac{1}{2}g(2t_0)^2 = 4\left(\frac{1}{2}gt_0^2\right) = 4h$.

Displacement during the second t_0 interval:

$$\Delta y_2 = y(2t_0) - y(t_0) = 4h - h = 3h$$

(b) Ball thrown vertically upward with initial speed v_0: Position equation: $y(t) = v_0 t - \frac{1}{2}gt^2$. After $t = t_0$, height is $h = v_0 t_0 - \frac{1}{2}gt_0^2 \implies v_0 t_0 = h + \frac{1}{2}gt_0^2$.
After time $t = 2t_0$, height h_2 is:

$$h_2 = v_0(2t_0) - \frac{1}{2}g(2t_0)^2 = 2(v_0 t_0) - 2gt_0^2 = 2\left(h + \frac{1}{2}gt_0^2\right) - 2gt_0^2 = 2h - gt_0^2$$

(c) Continuous dropping from building height $H = 12$ m: Let T be the regular time interval between drops. When ball 3 is released ($t = 0$), ball 2 has fallen for time T, and ball 1 strikes the ground after time $2T$.
Total fall distance for ball 1:

$$H = \frac{1}{2}g(2T)^2 = 2gT^2 = 12 \text{ m} \implies gT^2 = 6 \text{ m}$$

Distance fallen by ball 2 (currently in the air):

$$y_2 = \frac{1}{2}gT^2 = \frac{1}{2}(6 \text{ m}) = 3 \text{ m}$$

Height of ball 2 above the ground:

$$h_2 = H - y_2 = 12 \text{ m} - 3 \text{ m} = 9 \text{ m}$$

SOLUTION 11 LEVEL II

Separation of Oppositely Thrown Stones

Define an upward vertical coordinate system with the origin $y = 0$ at the cliff edge. Taking $v_0 = 6$ m/s and $g = 9.8$ m/s^2:

- Position of Stone A (thrown upward): $y_A(t) = +v_0 t - \frac{1}{2}gt^2$
- Position of Stone B (thrown downward): $y_B(t) = -v_0 t - \frac{1}{2}gt^2$

The separation distance $\Delta y(t)$ between the two stones is:

$$\Delta y(t) = y_A(t) - y_B(t) = \left(v_0 t - \frac{1}{2}gt^2\right) - \left(-v_0 t - \frac{1}{2}gt^2\right) = 2v_0 t$$

Notice that the gravitational acceleration term $-\frac{1}{2}gt^2$ cancels out completely.
At $t = 3$ s:

$$\Delta y = 2(6 \text{ m/s})(3 \text{ s}) = 36 \text{ m}$$

SOLUTION 12 **LEVEL II**

Collision of Two Counter-Moving Balls

Coordinate origin $y = 0$ at ground level, upward positive, $g = 10$ m/s^2. Positions as functions of time t:

$$y_1(t) = v_1 t - \frac{1}{2} g t^2 = 20t - 5t^2$$

$$y_2(t) = H - v_2 t - \frac{1}{2} g t^2 = 150 - 30t - 5t^2$$

(a) Collision time and height: Set $y_1(t_{\text{coll}}) = y_2(t_{\text{coll}})$:

$$20t_{\text{coll}} - 5t_{\text{coll}}^2 = 150 - 30t_{\text{coll}} - 5t_{\text{coll}}^2 \implies 50t_{\text{coll}} = 150 \implies t_{\text{coll}} = 3.0 \text{ s}$$

Substituting $t_{\text{coll}} = 3.0$ s into $y_1(t)$:

$$y_{\text{coll}} = 20(3) - 5(3)^2 = 60 - 45 = 15 \text{ m}$$

(b) Order of events: Time for Ball 1 to reach maximum height:

$$t_{\text{apex}} = \frac{v_1}{g} = \frac{20}{10} = 2.0 \text{ s}$$

Since $t_{\text{apex}} = 2.0$ s $< t_{\text{coll}} = 3.0$ s, **Ball 1 reaches its maximum height first** (and is moving downward when the collision occurs).

(c) Total distance traveled by Ball 1: Maximum height reached by Ball 1 above ground:

$$y_{\max} = \frac{v_1^2}{2g} = \frac{20^2}{2(10)} = 20 \text{ m}$$

Ball 1 travels 20 m upward to its apex, then falls from 20 m to the collision height of 15 m (a distance of 5 m).

$$d_{\text{total}} = 20 \text{ m} + (20 - 15) \text{ m} = 25 \text{ m}$$

1.5 Projectile Motion

SOLUTION 13 **LEVEL II**

Diverging Projectiles Distance

Both particles experience identical downward gravitational acceleration $\vec{g} = -g\hat{j}$. Thus, the relative acceleration between them is zero: $\vec{a}_{\text{rel}} = \vec{g} - \vec{g} = 0$.
The relative motion is pure uniform linear motion with relative velocity $\vec{v}_{\text{rel}} = \vec{v}_2 - \vec{v}_1$:

$$\vec{v}_1 = 0\hat{i} + v_0\hat{j}$$

$$\vec{v}_2 = (v_0 \cos 60^\circ)\hat{i} + (v_0 \sin 60^\circ)\hat{j} = \frac{1}{2}v_0\hat{i} + \frac{\sqrt{3}}{2}v_0\hat{j}$$

$$\vec{v}_{\text{rel}} = \frac{1}{2}v_0\hat{i} + \left(\frac{\sqrt{3}}{2} - 1\right)v_0\hat{j}$$

Calculating the magnitude of relative velocity $|\vec{v}_{\text{rel}}|$:

$$|\vec{v}_{\text{rel}}|^2 = v_0^2\left[\frac{1}{4} + \left(\frac{\sqrt{3}}{2} - 1\right)^2\right] = v_0^2\left[\frac{1}{4} + \frac{3}{4} - \sqrt{3} + 1\right] = v_0^2(2 - \sqrt{3})$$

$$|\vec{v}_{\text{rel}}| = v_0\sqrt{2 - \sqrt{3}}$$

With $v_0 = 25$ m/s and $t = 1.70$ s:

$d(t) = |\vec{v}_{\text{rel}}| \cdot t = 25 \times \sqrt{2 - 1.73205} \times 1.70 = 25 \times \sqrt{0.26795} \times 1.70 \approx 25 \times 0.5176 \times 1.70 \approx 22.0\,\text{m}$

SOLUTION 14

LEVEL II

Direct Aim at a Vertical Wall

With launch angle $\theta = 30^\circ$, initial speed $v_0 = 40$ m/s, distance $d = 70$ m, and $g = 9.8$ m/s^2:
1. Intended target height along the straight line of sight:

$$y_{\text{target}} = d\tan\theta = 70\tan 30^\circ = \frac{70}{\sqrt{3}} \approx 40.415\ \text{m}$$

2. Time of flight to reach the wall at distance d:

$$t = \frac{d}{v_0\cos\theta} = \frac{70}{40\cos 30^\circ} = \frac{70}{20\sqrt{3}} = \frac{3.5}{\sqrt{3}} \approx 2.0207\ \text{s}$$

3. Actual impact height on the wall:

$$y_{\text{impact}} = (v_0\sin\theta)t - \frac{1}{2}gt^2 = (d\tan\theta) - \frac{1}{2}gt^2 = y_{\text{target}} - \frac{1}{2}gt^2$$

4. Vertical deflection below the target:

$$\Delta y = y_{\text{target}} - y_{\text{impact}} = \frac{1}{2}gt^2 = \frac{1}{2}(9.8)\left(\frac{3.5}{\sqrt{3}}\right)^2 = 4.9 \times \frac{12.25}{3} \approx 20.0\ \text{m}$$

SOLUTION 15 LEVEL II

Speed versus Height Graph Analysis

Using the 1D kinematic relationship $v^2 = v_0^2 - 2g(y - y_0)$ between points A and B:

$$v_B^2 = v_A^2 - 2g(y_B - y_A) = v_A^2 - 2gd$$

Given $v_B = \frac{1}{3}v_A$ and $d = 0.40$ m, with $g = 9.8$ m/s^2:

$$\left(\frac{1}{3}v_A\right)^2 = v_A^2 - 2gd \implies \frac{1}{9}v_A^2 = v_A^2 - 2gd$$

$$\frac{8}{9}v_A^2 = 2gd \implies v_A^2 = \frac{9}{4}gd$$

$$v_A = \frac{3}{2}\sqrt{gd} = 1.5 \times \sqrt{9.8 \times 0.40} = 1.5 \times \sqrt{3.92} \approx 1.5 \times 1.9799 = 2.97 \text{ m/s}$$

SOLUTION 16 LEVEL II

Depth of a Water Well from Sound

Taking $g = 9.80$ m/s^2, speed of sound $v_s = 336$ m/s, and total time $T = 2.40$ s:

(a) Exact depth h: Total time $T = t_{\text{fall}} + t_{\text{sound}}$, where $t_{\text{fall}} = \sqrt{\frac{2h}{g}}$ and $t_{\text{sound}} = \frac{h}{v_s}$.

$$\sqrt{\frac{2h}{9.8}} + \frac{h}{336} = 2.40$$

Let $x = \sqrt{h}$:

$$\frac{x}{336}x^2 + \sqrt{\frac{2}{9.8}}x - 2.40 = 0 \implies \frac{1}{336}x^2 + 0.45175x - 2.40 = 0$$

Solving quadratic equation $x^2 + 151.79x - 806.4 = 0$:

$$x = \frac{-151.79 + \sqrt{151.79^2 + 4(806.4)}}{2} = \frac{-151.79 + \sqrt{23040.2 + 3225.6}}{2} \approx 5.122 \text{ m}^{1/2}$$

$$h = x^2 = (5.122)^2 \approx 26.24 \text{ m}$$

(b) Percentage error if sound speed is assumed infinite: Approximated depth h_{approx}:

$$h_{\text{approx}} = \frac{1}{2}gT^2 = \frac{1}{2}(9.80)(2.40)^2 = 28.22 \text{ m}$$

Percentage error:

$$\text{Error} = \frac{|h_{\text{approx}} - h|}{h} \times 100\% = \frac{|28.22 - 26.24|}{26.24} \times 100\% \approx 7.5\%$$

SOLUTION 17

LEVEL II

Dive-Bomber Release Angle

Plane velocity $v_0 = 280$ m/s at angle θ below horizontal. Altitude $y = 2150$ m, displacement magnitude $d = 3250$ m.

1. Horizontal displacement x:

$$x = \sqrt{d^2 - y^2} = \sqrt{3250^2 - 2150^2} = \sqrt{10562500 - 4622500} = \sqrt{5940000} \approx 2437.2 \text{ m}$$

2. Equations of motion (taking $g = 9.8$ m/s^2):

$$x = (v_0 \cos\theta)t \implies t = \frac{x}{v_0 \cos\theta}$$

$$y = (v_0 \sin\theta)t + \frac{1}{2}gt^2 = x\tan\theta + \frac{gx^2}{2v_0^2\cos^2\theta} = x\tan\theta + \frac{gx^2}{2v_0^2}(1 + \tan^2\theta)$$

Substitute numerical values:

$$\frac{gx^2}{2v_0^2} = \frac{9.8 \times (2437.2)^2}{2 \times 280^2} = \frac{58210000}{156800} \approx 371.24$$

$$2150 = 2437.2\tan\theta + 371.24(1 + \tan^2\theta)$$

$$371.24\tan^2\theta + 2437.2\tan\theta - 1778.76 = 0$$

Solving for $\tan\theta$:

$$\tan\theta = \frac{-2437.2 + \sqrt{2437.2^2 - 4(371.24)(-1778.76)}}{2(371.24)} = \frac{-2437.2 + 2930.5}{742.48} \approx 0.6644$$

$$\theta = \arctan(0.6644) \approx 33.6^\circ$$

SOLUTION 18

LEVEL II

Maximum Height to Range Ratio

(a) Derivation of H/R: Maximum height $H = \frac{v_0^2 \sin^2\theta}{2g}$. Horizontal range $R = \frac{v_0^2 \sin(2\theta)}{g} = \frac{2v_0^2 \sin\theta\cos\theta}{g}$. Dividing H by R:

$$\frac{H}{R} = \frac{\frac{v_0^2 \sin^2\theta}{2g}}{\frac{2v_0^2 \sin\theta\cos\theta}{g}} = \frac{\sin^2\theta}{4\sin\theta\cos\theta} = \frac{1}{4}\tan\theta$$

(b) Angle for $R = 2H$: Setting $R = 2H \implies \frac{H}{R} = \frac{1}{2}$:

$$\frac{1}{4}\tan\theta = \frac{1}{2} \implies \tan\theta = 2 \implies \theta = \arctan(2) \approx 63.4^\circ$$

SOLUTION 19

LEVEL II

Fixed Initial Speed with Variable Launch Angle

The range formula is $R(\theta) = \frac{v_0^2 \sin(2\theta)}{g}$.
The maximum horizontal range L for a given speed v_0 occurs at $\theta = 45^\circ$, where $\sin(90^\circ) = 1$:

$$L = \frac{v_0^2}{g}$$

For launch angle $\theta = 30^\circ$:

$$R(30^\circ) = \frac{v_0^2 \sin(2\times 30^\circ)}{g} = \frac{v_0^2 \sin(60^\circ)}{g} = L\sin(60^\circ) = \frac{\sqrt{3}}{2}L \approx 0.866L$$

SOLUTION 20

LEVEL II

Level-Ground Projectile with Range 60m

Given range $R = 60$ m, launch angle $\theta = 45^\circ$, and taking $g = 9.8$ m/s^2:

(a) Initial speed v_0:

$$R = \frac{v_0^2 \sin(2\theta)}{g} \implies 60 = \frac{v_0^2 \sin(90^\circ)}{9.8} \implies v_0^2 = 588 \implies v_0 = \sqrt{588} \approx 24.25 \text{ m/s}$$

(b) Time of flight T:

$$T = \frac{2v_0 \sin\theta}{g} = \frac{2(24.25)\sin 45^\circ}{9.8} = \frac{34.29}{9.8} \approx 3.50 \text{ s}$$

(c) Behavior for $\theta > 45^\circ$ with $R = 60$ m remaining constant: Since $R = \frac{v_0^2 \sin(2\theta)}{g}$ and $\sin(2\theta) < 1$ for $\theta > 45^\circ$, a **larger initial speed v_0** is required to achieve the same horizontal range. Because both v_0 and $\sin\theta$ increase, the total time of flight $T = \frac{2v_0 \sin\theta}{g}$ **increases significantly**. Physically, the projectile reaches a greater maximum height, spending more time in the air.

SOLUTION 21 LEVEL II

Impact Velocity and Vector Direction

Initial velocity vector: $\vec{v}_i = (v_i \cos\theta_i)\hat{i} + (v_i \sin\theta_i)\hat{j}$. On level ground, final velocity vector: $\vec{v}_f = (v_i \cos\theta_i)\hat{i} - (v_i \sin\theta_i)\hat{j}$.

(a) Angle between $\vec{v}_i$ and $\vec{v}_f$: The vector $\vec{v}_i$ points at angle $+\theta_i$ above the horizontal, and $\vec{v}_f$ points at angle $-\theta_i$ below the horizontal. The total angle between them is $2\theta_i$ (or $360^\circ - 2\theta_i$).

(b) Sum of velocity vectors $\vec{v}_i + \vec{v}_f$:

$$\vec{v}_i + \vec{v}_f = \left[(v_i \cos\theta_i)\hat{i} + (v_i \sin\theta_i)\hat{j}\right] + \left[(v_i \cos\theta_i)\hat{i} - (v_i \sin\theta_i)\hat{j}\right] = (2v_i \cos\theta_i)\hat{i}$$

SOLUTION 22 LEVEL III

Ramp Collision Sequence

Set up a coordinate frame aligned with the ramp inclined at angle α to the horizontal: x-axis directed up the incline, y-axis perpendicular to the ramp pointing outward. Gravity components along axes: $a_x = -g\sin\alpha$, $a_y = -g\cos\alpha$. Horizontal velocity v_0 before first collision resolves into:

$$v_{x0} = v_0 \cos\alpha, \quad v_{y0} = v_0 \sin\alpha \quad \text{(after elastic bounce)}$$

(a) Ramp angle α: Time for each bounce along y: $T = \frac{2v_{y0}}{g\cos\alpha} = \frac{2v_0 \sin\alpha}{g\cos\alpha} = \frac{2v_0}{g}\tan\alpha$. Since collisions are elastic, time between all consecutive bounces is identical: $\Delta t = T$. Total time to $(N+1)^{\text{st}}$ collision: $t_N = N \cdot T = \frac{2Nv_0 \tan\alpha}{g}$.

At $(N+1)^{\text{st}}$ collision, motion is perpendicular to ramp $\implies v_x(t_N) = 0$:

$$v_x(t_N) = v_{x0} + a_x t_N = v_0 \cos\alpha - (g\sin\alpha)\left(\frac{2Nv_0\tan\alpha}{g}\right) = 0$$

$$v_0\cos\alpha = 2Nv_0\sin\alpha\tan\alpha \implies \cos\alpha = 2N\frac{\sin^2\alpha}{\cos\alpha} \implies \tan^2\alpha = \frac{1}{2N}$$

$$\alpha = \arctan\left(\frac{1}{\sqrt{2N}}\right)$$

(b) Distance along ramp: Distance along ramp $x(t_N)$:

$$x(t_N) = v_{x0}t_N + \frac{1}{2}a_x t_N^2 = (v_0\cos\alpha)t_N - \frac{1}{2}(g\sin\alpha)t_N^2$$

Since $g\sin\alpha t_N = v_0\cos\alpha$:

$$x(t_N) = (v_0\cos\alpha)t_N - \frac{1}{2}(v_0\cos\alpha)t_N = \frac{1}{2}v_0\cos\alpha t_N = \frac{1}{2}v_0\cos\alpha\left(\frac{2Nv_0\tan\alpha}{g}\right) = \frac{Nv_0^2\sin\alpha}{g}$$

Since $\tan\alpha = \frac{1}{\sqrt{2N}} \implies \sin\alpha = \frac{1}{\sqrt{2N+1}}$:

$$d = \frac{Nv_0^2}{g\sqrt{2N+1}}$$

SOLUTION 23 LEVEL III

Tennis Ball Launcher on an Accelerating Train

Work in the non-inertial frame of reference of the accelerating train. Effective horizontal acceleration pushing backward relative to train: $a_{\text{rel}} = 2.00\ \text{m/s}^2$. Vertical acceleration remains downward gravity: $g = 9.80\ \text{m/s}^2$. Initial launch velocity $v_0 = 25.0\ \text{m/s}$, max height $H_{\max} = 10.0\ \text{m}$.

(a) Launch angle θ_0: Vertical velocity component v_{y0}:

$$v_{y0}^2 = 2gH_{\max} = 2(9.80)(10.0) = 196 \implies v_{y0} = 14.0\ \text{m/s}$$

Since $v_{y0} = v_0\sin\theta_0$:

$$\sin\theta_0 = \frac{v_{y0}}{v_0} = \frac{14.0}{25.0} = 0.560 \implies \theta_0 = \arcsin(0.560) \approx 34.1^\circ$$

(b) Landing distance from front end: Total time in air T:

$$T = \frac{2v_{y0}}{g} = \frac{2(14.0)}{9.80} = 2.857\ \text{s}$$

Horizontal initial velocity relative to train:

$$v_{x0} = v_0 \cos\theta_0 = 25.0\cos(34.06^\circ) \approx 20.71 \text{ m/s}$$

Relative horizontal distance traveled relative to the train floor:

$$x_{\text{rel}} = v_{x0}T + \frac{1}{2}a_{\text{rel}}T^2 = (20.71)(2.857) + \frac{1}{2}(2.00)(2.857)^2 = 59.17 + 8.16 = 67.33 \text{ m}$$

The ball lands 67.3 m toward the rear from the front end of the train.

SOLUTION 24

LEVEL III

Cliff-Top Projectiles at Opposite Angles

Height $h = 40$ m, $v_0 = 30$ m/s, angle $\theta = 30^\circ$, $g = 9.8$ m/s^2.

(a) Difference in time of flight Δt: Ball 1 is launched upward at $+30^\circ$. It ascends, reaches its peak, and descends back to the cliff height ($y = 0$) moving downward at speed v_0 at angle -30°. This state matches the initial launch of Ball 2. Therefore, Δt is the time Ball 1 spends executing level-ground motion above the cliff height:

$$\Delta t = \frac{2v_0 \sin\theta}{g} = \frac{2(30)\sin 30^\circ}{9.8} = \frac{30}{9.8} \approx 3.06 \text{ s}$$

(b) Horizontal separation distance on the ground: Since both balls have the exact same horizontal velocity $v_x = v_0 \cos 30^\circ = 15\sqrt{3}$ m/s, the horizontal separation Δx upon landing is:

$$\Delta x = v_x \cdot \Delta t = (15\sqrt{3})(3.0612) \approx 79.5 \text{ m}$$

(c) Physical Observation: The time difference $\Delta t = 3.06$ s and distance $\Delta x = 79.5$ m are **identical** to the flight time and range of a level-ground projectile launched at $v_0 = 30$ m/s at $\theta = 30^\circ$. This confirms that gravity acts symmetrically on both trajectories above the cliff plane.

SOLUTION 25

LEVEL III

Vertical Component of Impact Velocity against a Wall

From the graph, $v_y(x)$ is a linear function of x. At $x = 0$, $v_{y0} = 5.0$ m/s. At $x = 10$ m (midpoint $x_0 = 3$ grid units = 10 m), $v_y = 0$.

1. Time taken to reach peak t_p (where $v_y = 0$, taking $g = 9.8\ \text{m/s}^2$):

$$v_y(t_p) = v_{y0} - g t_p = 0 \implies t_p = \frac{v_{y0}}{g} = \frac{5.0}{9.8} \approx 0.5102\ \text{s}$$

2. Horizontal distance to peak $x(t_p) = 10$ m:

$$x(t_p) = v_x \cdot t_p \implies 10 = v_x(0.5102) \implies v_x = \frac{10}{0.5102} = 19.6\ \text{m/s}$$

3. Launch angle θ_0:

$$\tan\theta_0 = \frac{v_{y0}}{v_x} = \frac{5.0}{19.6} \approx 0.2551 \implies \theta_0 = \arctan(0.2551) \approx 14.3^\circ$$